

$$yS = (y \times 1)^\#(S)$$

by

§1.616 — pre-composing a relation by a map is an inverse image

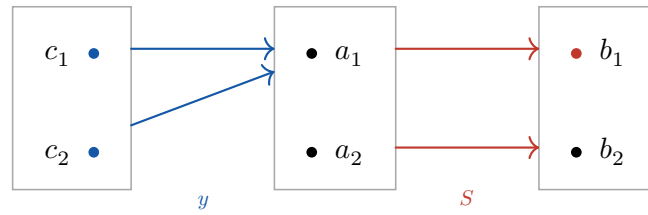
For a **map** $y : C \rightarrow A$ and a **relation** $S : A \rightarrow B$, the composite $yS : C \rightarrow B$, as a subobject of $C \times B$, is the inverse image of S along $y \times 1$:

$$yS = (y \times 1)^\#(S)$$

where $y \times 1 = y \times 1_B : C \times B \rightarrow A \times B$, $(c, b) \mapsto (y(c), b)$. As a pullback:

$$\begin{array}{ccc} C \times B & \xrightarrow{y \times 1} & A \times B \\ \uparrow \text{green} & & \uparrow \text{red} \\ yS & \xleftarrow{(y \times 1)^\#} & S \end{array}$$

§0.1. In SET



As a composite: $c (yS) b \iff (y(c), b) \in S$. Both $c_1, c_2 \mapsto a_1$ and $a_1 S b_1$, so

$$yS = \{(c_1, b_1), (c_2, b_1)\} \quad (c_2 \rightarrow a_1 \rightarrow b_1).$$

As an inverse image: $y \times 1$ sends $(c, b) \mapsto (a_1, b)$, so the preimage of S is

$$(y \times 1)^\#(S) = \{(c, b) : (y(c), b) \in S\} = \{(c_1, b_1), (c_2, b_1)\}.$$

Same subset of $C \times B$.

★ Punchline

Pre-composing S by the map y is pulling S back along $y \times 1$. So inverse images preserving unions gives, for free, $y(S \cup T) = yS \cup yT$.